Day 017 | Unit 02 | Teacher plan

Positive Rational Numbers: Fractions and Mixed Numbers

Multiply Mixed Numbers: Readiness

Standards: 6.NS.A.1

Learning target: I can use fraction multiplication to prepare for fraction division.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. For a product of fractions, multiply numerators and multiply denominators, then simplify. Convert mixed numbers to improper fractions first. Multiplying by a fraction less than 1 makes a positive quantity smaller.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 15

Day 017 | Unit 02 | Lesson

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01 / Learn

For a product of fractions, multiply numerators and multiply denominators, then simplify. Convert mixed numbers to improper fractions first. Multiplying by a fraction less than 1 makes a positive quantity smaller.

Worked example 1

Calculate 2 1/4 × 2 1/4. Simplify.

1. Rename the numbers as fractions: 2 1/4 = 9/4 and 2 1/4 = 9/4.

2. Keep multiplication and multiply the fractions: 9/4 × 9/4.

3. Multiply numerators and denominators: (9 × 9) / (4 × 4) = 81/16.

4. Simplify by dividing top and bottom by 1: 81/16.

5. Check the result: The product of 9/4 and 9/4 is 81/16. A positive factor below 1 makes the other positive factor smaller.

Answer: 81/16

Worked example 2

Calculate 3 1/4 × 1 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 31/8
B. 39/8
C. 8/39
D. 47/8

1. For a product of fractions, multiply numerators and multiply denominators, then simplify.

2. 13/4 × 3/2 = 39/8.

Answer: B - 39/8

Day 017 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1 1/4 × 2 1/2. Simplify.

02 / Constructed response

Calculate 2 1/2 × 1 1/3. Simplify.

03 / Constructed response

Calculate 3 1/2 × 1 1/2. Simplify.

04 / Constructed response

Calculate 3 3/4 × 1 1/3. Simplify.

05 / Constructed response

Calculate 3 1/2 × 1 1/3. Simplify.

06 / Constructed response

Calculate 1 1/2 × 1 1/2. Simplify.

07 / Constructed response

Calculate 2 1/2 × 1 1/3. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 3 3/4 × 1 1/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 6
B. 4
C. 1/5
D. 5

09 / Multiple choice

Calculate 1 1/2 × 1 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 4/9
B. 9/4
C. 5/4
D. 13/4

10 / Constructed response

Calculate 3 1/4 × 1 1/3. Simplify and explain why your answer is reasonable.

Day 017 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

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1. 25/8

5/4 × 5/2 = 25/8.

6.NS.A.1 | Constructed response

2. 10/3

5/2 × 4/3 = 10/3.

6.NS.A.1 | Constructed response

3. 21/4

7/2 × 3/2 = 21/4.

6.NS.A.1 | Constructed response

4. 5

15/4 × 4/3 = 5.

6.NS.A.1 | Constructed response

5. 14/3

7/2 × 4/3 = 14/3.

6.NS.A.1 | Constructed response

6. 9/4

3/2 × 3/2 = 9/4.

6.NS.A.1 | Constructed response

7. 10/3

5/2 × 4/3 = 10/3.

6.NS.A.1 | Constructed response

8. D - 5

15/4 × 4/3 = 5.

6.NS.A.1 | Multiple choice

9. B - 9/4

3/2 × 3/2 = 9/4.

6.NS.A.1 | Multiple choice

10. 13/3

13/4 × 4/3 = 13/3.

6.NS.A.1 | Constructed response